Ira A. Fulton Schools of Engineering
Arizona State University

CSE 579: Knowledge Representation and Reasoning

Simple Answer Set Programming

Purpose

The primary objective of this assignment is to showcase your mastery of simple Answer Set Programming using Clingo. By successfully solving problems 1 through 9, you will not only gain hands-on experience in writing Answer Set Programs but also become adept at utilizing Clingo to find solutions.

Objectives

Learners will be able to:

- Demonstrate Proficiency in Simple Answer Set Programming with Clingo
- Execute and Document Clingo Commands Locally

Technology Requirements

- Clingo - https://github.com/zhunyoung/clingo

Project Description

To complete the assignment, you will use Clingo to solve nine introductory-level problems. You will learn to write simple answer set programs to solve problems and get to know the language of Clingo. Full instructions and assignment details follow.

NOTE: As the problems are drawn from this module's lectures, you will find the rules that are necessary to follow in the lecture slides. The lectures also include Dr. Lee's walk-throughs of the solutions to most of these problems.

Directions

1. Download the assignment template and the problems. Read all of the problems and review the evaluation criteria section that follows. NOTE: All of the problems are provided in the attached PDF for initial review and reference as you work in clingo.

a. CSE579_Simple Answer Set Programming_Template
b. CSE579_Simple Answer Set Programming_Problems

2. Access clingo on your local machine.

    a. You must use clingo to complete this assignment. You also must complete this assignment using clingo on your **local machine**, because the online version of the tool will not include the command line, which you are required to submit to complete the assignment. The installation guide is available at https://github.com/zhunyoung/clingo.

3. Complete the problems.

    a. Complete problems 1 through 9 and use the attached template (provided above) to collect the information required for each problem. Note that some hints and clingo rules are also given in the template to help you to complete these 9 problems. Once you have filled in all the tables in the template, save it as a pdf file named "pa1.pdf".

Submission Directions for Project Deliverables

You are given an unlimited number of attempts to submit your best work. The number of attempts is given to anticipate any submission errors you may have in regards to properly submitting your best work within the deadline (e.g., accidentally submitting the wrong paper). It is not meant for you to receive multiple rounds of feedback and then one (1) final submission. Only your most recent submission will be assessed.

You must submit your Simple Answer Set Programming deliverable in this submission space. Learners may not email or use other means to submit any assignment or project for review, including feedback, and grading.

The Simple Answer Set Programming includes one (1) deliverable:

- **Answer Set PDF:** You must submit your assignment as a single PDF, using the provided template: Template_CSE 579_Simple Answer Set Programming Assignment. Your file upload must follow the correct naming convention: Your Name_CSE 579_Simple Answer Set Programming Assignment.

Your submission will be reviewed by the course team before finalizing your assignment grade. To submit your work follow the directions below:

1. In your course, go to **Submission: Simple Answer Set Programming**.

2. Click **Start Assignment**.
3. Click **Choose File**.
4. Locate and select **one (1)** deliverable file from your device.
5. If needed, click **+Add Another File** and repeat Steps 3 and 4 until all deliverables are added.
6. Select the **agreement** and then click **Submit Assignment**.
7. (If needed and allowed) To resubmit files:
    a. Return to the Canvas submission space, click **New Attempt**, and repeat the process from Step 3.

Evaluation

This assignment is staff-graded. There are a total of 9 problems for a total of 100 points.

Rubric

Please review the rubric found on the submission page for this assignment.